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Learning about ventilators

A ventilator is a machine that breathes for you or helps you breathe. It is also called a breathing machine or respirator. The ventilator:

- Is a computerized machine with knobs and buttons that are controlled by a respiratory therapist, nurse, or health care provider.
- Has tubes that connect to the person through a breathing tube. The breathing tube is placed in the person's mouth or in an opening through the neck into the windpipe (trachea). This opening is called a tracheostomy. It is often needed for those who have to be on the ventilator for a longer period of time.
- May sometimes be connected to a tight fitting mask over the mouth or nose of the person (similar to CPAP and BiPAP).
- Makes noise and has alarms that alert the health care team when something needs to be fixed or changed.

How Does Being on a Ventilator Feel?

A person receives medicine to remain comfortable while on a ventilator, especially if they have a breathing tube in their mouth. The medicine may cause people to be too sleepy to open their eyes or stay awake for more than a few minutes.

People cannot talk because of the breathing tube. When they are awake enough to open their eyes and move, they can communicate in writing and sometimes by lip reading.

People on ventilators will have many wires and tubes on them. This may look scary, but these wires and tubes help to carefully monitor them.

Some people may have restraints. These are used to prevent them from pulling out any important tubes and wires.

Why are Ventilators Needed?

People are placed on ventilators when they cannot breathe adequately on their own. This may be for any of the following reasons:

- To make sure the person is getting enough oxygen and is getting rid of carbon dioxide.
- After surgery, people may need a ventilator to breathe for them when they have had medicine that causes them to be sleepy and their breathing has not returned to normal.
- A person has an illness or injury and is not able to breathe normally.
- A person isn't awake enough to be able to cough on their own to clear secretions from their airways.

Most of the time, a ventilator is needed only for a short time -- hours, days, or weeks. But in some cases, a ventilator is needed for months, years, or indefinitely.

What Kind of Care Does a Person Need When on a Ventilator?

In the hospital, a person on a ventilator is watched closely by their health care team, including providers, nurses, and respiratory therapists.

People who need ventilators for long periods may stay in long-term care facilities. Some people with a tracheostomy may be able to be at home, but they do require a lot of support from family or other healthcare professionals.

People on a ventilator are watched carefully for lung infections. When connected to a ventilator, a person has a hard time coughing out mucus. If mucus collects, the lungs do not get enough oxygen. The mucus can also lead to pneumonia. To get rid of the mucus, a procedure called suctioning is needed. This is done by inserting a small thin tube into the person's mouth or neck opening to vacuum out the mucus.

When the ventilator is used for more than a few days, the person may receive nutrition through tubes into either a vein or their stomach.

Because the person cannot speak, special efforts need to be made to monitor them and provide them with other ways to communicate.

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